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University of Kentucky Researchers Develop “Bring Your Own Projector” Machine Vision Camera

A single board pairs a conventional industrial camera with a built-in HDMI controller, turning any projector into a structured-light 3D scanner — and chaining camera to camera over the video signal itself.

LEXINGTON, Ky. — Researchers at the University of Kentucky have introduced a machine vision camera that incorporates an HDMI controller for interfacing directly to commodity projectors. The board is a traditional high-speed machine vision camera — a global-shutter sensor streaming full frames over USB 3.0 — with a complete HDMI receiver and transmitter integrated alongside it on the same FPGA. Video from a host computer enters the camera and passes through to a projector. Because the camera sits *in* the video path rather than beside it, it knows precisely what is being projected, and when.

The team calls the result BYOP, for *Bring Your Own Projector*: connect any HDMI projector and the system becomes a structured-light 3D scanner, with no external trigger box, frame grabber, or custom synchronization cabling. The reason to want that is access. Structured-light systems are conventionally built around dedicated projection light engines — metrology modules made in small volumes, at prices that dominate the cost of the instrument. To an industrial buyer specifying a measuring machine, that cost is simply part of the machine. To everyone else it is a barrier, and it has confined structured-light 3D scanning to the groups that can afford the light engine.

Academic laboratories, this one included, have long worked the other way round — adapting to the commodity projectors they can actually buy. Those projectors are mass-produced against the 3D-Ready and DLP market, and they bring real advantages of their own: high luminance, a wide choice of fields of view and throw distances, and, with LED illumination, an operating life measured in tens of thousands of hours. What has made them awkward to build with was never their optics — it was the difficulty of making a camera keep time with a device never designed to be part of a measuring instrument.

The instrument is built as a stack. Three of its boards are commercially available parts from **Alchitry** — the Pt V2 mainboard carrying the Artix-7 FPGA, the Ft+ for the USB 3.0 link, and the Hd+ for HDMI in and out — and on top of them sits a **custom-designed sensor board**, developed for this project, carrying the PYTHON 1300 and its power supplies. The boards mate through a common high-density stacking connector, so the whole camera assembles without a single internal cable and drops into the printed enclosure shown in Figure 1. Table 1 summarises the hardware.

Why put HDMI inside a camera?

Structured-light 3D scanning works by projecting a sequence of patterns onto an object and photographing each one. The measurement is only as good as the synchronization: the camera must expose once per projected pattern, at a known moment within that pattern's display time. Conventionally, that means a trigger generator, a cable harness, and a calibration step for the delay between them — an assembly that has to be reworked whenever the projector, the frame rate, or the pattern set changes. Integrating the video interface into the camera removes the problem rather than managing it. The projected frames and the camera's exposures come from the same signal, inside one device.

It also solves a subtler problem. A PC's graphics hardware is not a real-time system: it may hold the same image on screen for an extra refresh, or deliver the next one late, and nothing in the video signal announces which has happened. The camera therefore reads the **top-left pixel** of every incoming frame and watches that value for change. A change means a genuinely new pattern has arrived and the frame should be captured; no change means the computer has repeated itself and the frame should be ignored. That sampled value then travels with the image: it is embedded in the **frame header** the camera sends to the host, so the PC can line the sequence of captured frames up against the sequence of patterns it played. The pattern sequence and the captured sequence stay in step even though the machine driving them makes no timing guarantees.

The imager

The camera is built around the **onsemi PYTHON 1300**, a global-shutter CMOS sensor with 1280×1024 resolution — 1.3 megapixels — reading out over four LVDS channels at 720 Mbit/s each. The sensor is socketed, and the same board accepts either the **monochrome** or the **colour** variant of the part. Its pixels are $4.8\mu\text{m}$ square, with a **global shutter**, and are delivered at **10 bits** of depth — 1024 grey levels rather than 256. At full resolution and 10-bit depth the sensor is capable of **210 frames per second**. For live streaming, the camera delivers **120 frames per second** over its USB-C connection, at the sensor's full 1280×1024 and full 10-bit depth — no cropping, no windowing, and no reduction in depth. For slower USB 2.0 connections, the camera does not have to stream at all. Frames are captured into **on-board DDR3 memory** as they are projected and downloaded afterwards, so the speed of the scan is set by the HDMI frame rate rather than by the streaming rate of the host connection.

No host HDMI source required

The camera does not have to be fed video at all. The board generates its own pattern sequences internally and drives the projector directly, at frame rates up to **120 per second** — the same rate at which it streams full frames. That rate is not arbitrary. **3D-Ready DLP projectors** — a large and inexpensive installed base, built to accept 120 Hz for shutter-glasses stereo — take a 120 Hz signal as a matter of course. The camera drives them at their native rate, and the highest standard mode it drives at 120 Hz is **800×600**. The entire structured-light loop — project a pattern, capture it, advance to the next — therefore runs on the camera itself, at 120 Hz, with nothing driving the video path but the camera. A computer is needed only to collect the resulting images, not to generate what is projected.

Where a host *is* driving the display, the FPGA can be configured to work either way. It can **replace** the incoming pixels with its own structured-light scan patterns, generated to suit the resolution arriving on the cable, and trigger the camera on every vertical sync — the host supplies the video timing, the camera supplies the patterns. Or it can **pass** the host's pixels through untouched and

use the top-left-pixel mechanism to decide when to trigger, leaving the pattern sequence entirely to the software driving the display. The same hardware serves both arrangements.

Any projector — and a host offered only what works

Nothing about the camera is tied to a particular projector, or to a particular projection technology. On power-up it reads the connected projector’s **EDID** — the data block every display serves back over the video cable, listing the resolutions and frame rates it will accept — and adapts to whatever it finds there. DLP, LCD or LCoS, the camera drives the projector at a mode that projector itself advertises, rather than at an assumed one. The 120 Hz 3D-Ready projectors described above are simply the fastest case, not a requirement.

When a host PC is in the video path, the camera goes a step further: it builds an EDID of its own and serves that to the computer, offering the projector’s own mode list narrowed to those modes the camera can also receive. The PC is therefore shown a set of display modes already known to work end to end, and the operator chooses among them in the ordinary display settings.

Nothing in that arrangement is specific to projectors, and the group expects to put it to work on high-speed gaming monitors next. Displays running at **200 to 210 Hz** are now commodity items, and at those rates a screen meets the PYTHON 1300 at the top of its own range — 210 frames per second. The application is *deflectometry*, where the pattern is reflected off a specular surface rather than projected onto a diffuse one, and where shiny parts that defeat a projector are exactly the ones worth measuring. That is intended future work, not a capability of the board today.

Daisy-chaining: one video signal, many cameras

Because the HDMI path runs *through* the camera — a separate input and output port, visible in Figure 2 — cameras can be chained. The host drives the first unit; each unit forwards the video onward to the next and, finally, to the projector. Every camera in the chain sees the same frames and can key to them.

That makes multi-camera geometries straightforward to assemble:

- **Active stereo** — two or more cameras viewing a projector-illuminated scene, capturing the same pattern from different viewpoints.
- **Multi-view capture** — cameras arranged around a subject, all keyed to a common projected sequence.
- **Extended baselines** — units placed far apart, coordinated over the same cable that already carries the video.

Background — the work this camera comes from

The architecture announced here was proposed and argued for in this group’s earlier published work. What is new is that it has been built, and measured. Readers who want the reasoning behind the design will find it in the following.

- M. P. Ruffner, Y. Yu and D. L. Lau, “Structured light smart camera for spatial augmented reality applications,” *Proc. SPIE* **10932**, Emerging Digital Micromirror Device Based Systems and Applications XI, 109320J (2019). [doi:10.1117/12.2508553](https://doi.org/10.1117/12.2508553)
The direct predecessor. It sets out why a PC cannot be trusted to synchronize a projector and a camera, and proposes putting the image sensor and an HDMI input and output onto the FPGA controller itself — the design this camera realizes.

- M. P. Ruffner, “Design of a Machine Vision Camera for Spatial Augmented Reality,” MSEE thesis, University of Kentucky (2018). uknowledge.uky.edu/ece_etds/129
The long-form treatment of the same design, including the power and layout work that a sensor board of this kind stands or falls on.
- Y. Yu, D. L. Lau, M. P. Ruffner and K. Liu, “Dual-projector structured light 3D shape measurement,” *Applied Optics* **59**(4), 964–974 (2020). [doi:10.1364/AO.378363](https://doi.org/10.1364/AO.378363)
Why more than one light engine is worth the trouble: fewer occlusions, better signal-to-noise, fewer frames per scan, and multi-path made visible. The argument that makes a chainable camera worth building.
- Y. Yu and D. L. Lau, “3D scanning by means of dual-projector structured light illumination,” *Proc. SPIE* **10932**, 109320M (2019). [doi:10.1117/12.2510990](https://doi.org/10.1117/12.2510990)
The conference treatment of the above, with the temporal-frequency multiplexing that lets two projectors run at once.

“This system is the culmination of years of research and development into structured light imaging, and by making commodity projector synchronization so easy, I hope it unlocks years more of technological improvements in SLI.”

— Daniel L. Lau, DataBeam Professor of Electrical and Computer Engineering, University of Kentucky

About

The BYOP camera comes out of the laboratory of **Daniel L. Lau**, DataBeam Professor of Electrical and Computer Engineering at the University of Kentucky, whose career in three-dimensional imaging spans twenty-five years. Since joining the University of Kentucky in 2001, he has worked on structured-light 3D imaging — both the algorithms that make real-time, high-accuracy shape measurement possible and the instruments that have to carry them — work that established non-contact 3D fingerprint capture as a practical alternative to contact scanning and that has reached industry through two companies he co-founded, FlashScan3D and Seikowave. He is also a pioneer of digital halftoning, and in 2026 was elected a Fellow of the IEEE for contributions to digital printing and 3D imaging.

Image sensor	onsemi PYTHON 1300, socketed — monochrome or colour
Resolution	1280 × 1024 (1.3 MP), global shutter
Pixel depth	10 bits per pixel (1024 grey levels); 8-bit selectable
Sensor frame rate	210 frames/s at full resolution, 10-bit
Streamed rate	120 frames/s sustained over USB-C
Camera timing	Free-running, or keyed to the projector — HDMI sync not required
Buffered capture	To on-board DDR3 at sensor rate, downloaded after
Host interface	USB 3.0, byte-exact — 325 MB/s achieved in testing
Video in / out	HDMI receive and transmit, pass-through
Projector support	Reads the projector's EDID; any projector in the window
Pass-through window	25–90 MHz pixel clock, enforced in the EDID served to the host
Standalone output	Board-generated video, no HDMI source needed
Highest 120 Hz mode	800×600 at 120 Hz (3D-Ready DLP projectors)
Verified modes	640 × 480 through 1280 × 800, 60 to 120 Hz
Processing	Xilinx Artix-7 FPGA with DDR3 frame buffer
Construction	Alchitry Pt V2, Ft+ and Hd+ boards plus a custom sensor board
On-board patterns	Sinusoidal fringe generation, radiometric linearization

Table 1: **BYOP camera — hardware summary.**



Figure 1: **The complete camera, opened.** The board stack sits in its printed enclosure: the HDMI board at the bottom, the USB 3.0 image link and the Artix-7 FPGA above it, and the PYTHON 1300 sensor board on top, its glass lid facing up. The 48-pin sensor was soldered by hand in the lab. At right, the enclosure's lid carries the lens and drops straight onto the stack. *Photo: University of Kentucky.*



Figure 2: **The connector edge.** The two HDMI ports at the bottom are the video input and output. The *input* is fed by the host PC, or by the previous camera when units are daisy-chained; the *output* drives the projector, or the next camera in the chain. Above them, the lower USB-C connector carries the image stream to the host. The sensor board is the topmost in the stack. *Photo: University of Kentucky.*